

1. The architectural complexity, while beneficial, may lead to **high computational cost and memory consumption**—a serious limitation in 3D volumetric processing.

2. No quantitative analysis of computational overhead (e.g., FLOPs, GPU usage, inference time) is provided.

This concern seems valid, as the architecture looks very complex at first glance but we due to our implementation of inception of the blocks the actual parameter and MAC count of the overall architecture is isn't nearly something to be concerned about.

Our model's Complexity:

- Trainable params: 4,008,868 (~4.01M)
- Total FLOPs: 201,153,318,912(~201.1G)

Compared To others(As reported in [this paper](#)[page 14]):

Methods	Parameters (M)	FLOPs (G)
3D U-Net	7.74	586.71
Attention U-Net	17.12	588.26
VNet	7.03	769.30
Swin-Unet	27.15	250.88
HDC-Net	0.29	25.62
TransUNet	105.18	1,205.76
TransBTS	30.61	263.82
MCSLF-Net	24.49	142.43

Our model might not be most efficient in terms of parameter count or flops, falling behind ultra lightweight models(Like [HDC-Net](#)), but it is comfortably in the category of lightweight models, having less parameters than a regular 3D U-Net.

As per GPU usage and Inference time, that would be hard to compare as that isn't reported in many paper. But for our model-

GPU VRAM Usage: ~11 GiB(For Batch Size of 2), ~13.5GiB(For a Batch Size of 3)

Average inference time per batch (2 volumes per batch): 0.8853 s

Average inference time per volume (batch Size=2): 0.4427 s

Average inference time per batch (3 volumes): 1.3295 s

Average inference time per volume (batch=3): 0.4432 s

*These are reported as per partial test runs on Google Colab, using a Tesla T4 GPU.

6. The absence of data augmentation strategies (rotation, scaling, flipping, intensity variation) weakens generalization claims.

Data augmentation strategies are usually very useful for 2D segmentation of 3D volumes. Where you take each 'slice' from a 3D Volume, treat it like a 2D data, run convolution over them with 2D 'windows' and stack them at the end when each slice has went through the same convolution process. You need data augmentation strategies like rotation, scaling, flipping, etc. because slicing up a 3D data means you loose the contextual information from other parts of the data. As we are using 3D convolutional 'windows' to segment a 3D data, the use of data augmentation strategies seems redundant as we don't suffer from same problem 'slice-by-slice' segmentation faces.

On top of that we have other means of bringing in contextual information in the network though multi-scale inputs, skip connections and attention gates. As a sum of all that strategies we are hardly loosing any contextual information.

*Though effects of data augmentation strategies can be tested if needed.

8. The training hyperparameters (learning rate, epochs, batch size, optimizer) are not reported, reducing reproducibility.

Learning Rate: 0.0001

Optimizer: Adam

Epochs: 150

Batch Size: 2

The Learning Rate and Optimizer was chosen as standard, used for most segmentation task. The number of epochs was chosen arbitrarily, we can easily run the model for more epochs if needed. Batch Size was chosen due to limitations on GPU VRAM.

4. The model's scalability to higher-resolution volumes or clinical datasets with noise and artifacts is not demonstrated.

5. The decision to crop data to 128^3 volumes risks losing peripheral tumor context, especially in large or irregular tumors.

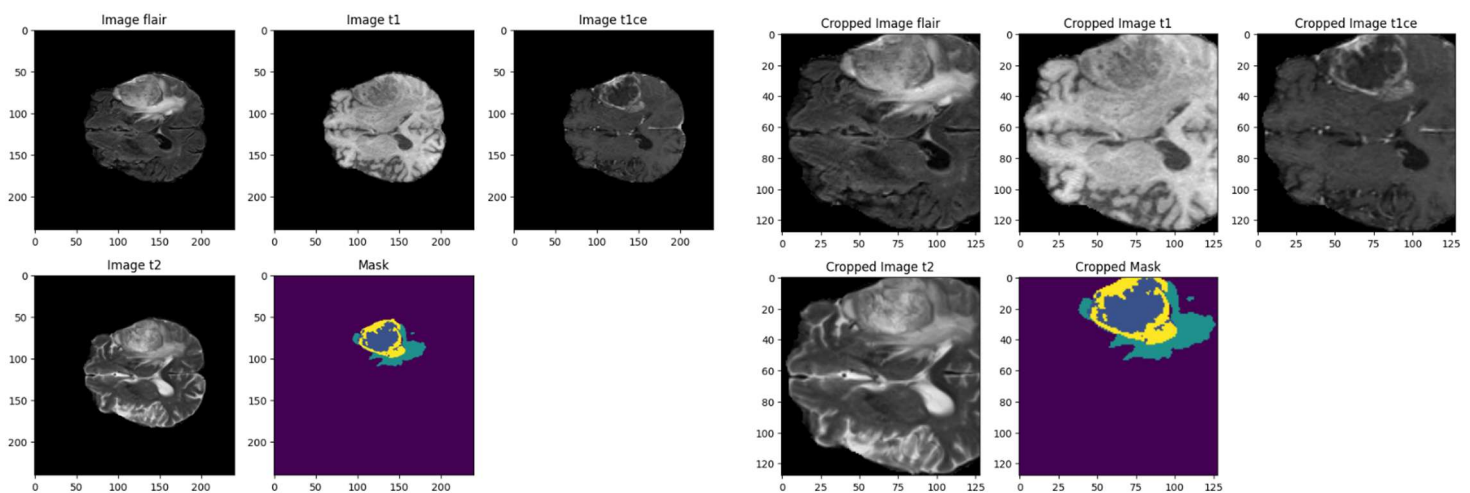
The reason we don't demonstrate model's scalability to be higher noise data is because medical imaging is hardly ever noisy. Medical images are most of the time captured in a very controlled environment mitigating their risk of them being noisy.

As for higher resolution volumes, most medical image datasets don't go above the size of 512^3 . The density of the model is theoretically capable enough to handle upto that size.

Though if needed we can easily test the model's scalability to that size. And if the model doesn't seem to handle that kind of resolution(which theoretically is very unlikely), we can easily increase the model depth(number of encoder-decoder stages) to make the model capable enough to handle higher resolutions.

Now for cropping it to a size of 128^3 ; when we have maxpooling operations involved in a network, it's good to have the data size in terms of 2^n .

Our lack of concern regarding it loosing peripheral context can be explained through the example below:



In this we are seeing exactly the middle slice of an example data. Being the middle slice this is the worse case scenario of how the crop effects our data, as the brain can't be any wider in any other portion. As even in the widest part of the brain more than 90% of the data is still present. Mean we can make a rough estimate that >97% of the useful data is still present. Through cropping we are discarding the black empty portion around the brain, making the date more light weight and easier to handle. We determined losing some of the brain in our training data would be an effective trade-off as even if we feed the model uncropped data afterwards when the training is done, it predicts the ground truth mask with effectively same accuracy.

9. There is no evidence that the hybrid loss was optimized or compared against other combinations (e.g., Tversky or Boundary loss).

The hybrid loss strategy was determined during the development phase through an theoretical understanding and preliminary testing, by which we deemed it useful and computationally effective enough to use it in the actual test run.

We don't have any recorded data of how it performs against dice or focal loss alone or any other commonly used complex losses like Tversky or Boundary loss. If found necessary we can do smaller partial runs to determine how they perform against each other.

11. No Dice Similarity Coefficient (DSC), Hausdorff Distance, or Precision–Recall curves are reported, which are standard for brain tumor segmentation.

12. The 99.03% accuracy may be misleading due to class imbalance — healthy voxels dominate MRI volumes.

13. No qualitative comparison (visual segmentation overlays) or quantitative benchmark comparison with existing models (e.g., Attention U-Net, 3D Dense U-Net, or V-Net) is provided.

We are aware that accuracy is not a good measure of performance in medical image segmentation tasks. The accuracy claim is stated as sort of an ‘hook’ in the beginning halves of the paper. We have recorded more standard matrices such as IoU, Dice Per class, Precision, Recall, throughout the run; all of which will be reported in the result section, along with qualitative comparisons with existing popular models.

15. The architecture integrates attention mechanisms, inherently offering some interpretability through activation maps. However, the paper does not present visual attention maps or feature saliency visualizations to confirm interpretability claims.

The attention gate mechanism is taken directly from the [Attention-UNet](#). The mechanism is widely explored and extensively studied. We didn’t include any extensive explanation in it’s working deeming it mostly ‘common knowledge’.

Although if we want to make interpretability claims, we’ll need to includes visual attention maps. The lack of interpretability claims is following the lack of attention maps.

10. The validation strategy (cross-validation vs. hold-out) is not described; without this, accuracy claims could be dataset-specific.

We used a 75-25 split without a test set. There are possibilities of this leading overfit to validation. But in our case 150 epochs proved to be an small enough run that we didn’t face that problem. That theory is supported by our loss curve, which showed a consistent downwards trend with no sudden falls or irregularities.

Though if time permits we can also test the effects of 3-fold cross-validation on the same dataset, or even some other validation strategies.

3. The DRIC block structure, though theoretically advantageous, lacks ablation studies proving the necessity of each subcomponent (Dense, Residual, and Inception) when combined.

The DRIC block structure was designed through studying various works that employ either Dense, Residual or Inception strategies to overcome different shortcomings of the U-Net architecture. There can be section on how each component of the block helps in what we aim to achieve by including that sub-component, alongside supporting literature that explores the benefits of each strategy.

But practically exploring if our theory stands sound would be almost futile. As testing and recoding the effect of each subcomponent behaves would greatly increase the scope of work.

7. The normalization approach, though standard, might not address MRI intensity non-standardization across scanners, an issue often mitigated by histogram matching or Z-score normalization

14. Lack of statistical validation (p-values or confidence intervals) limits the credibility of results.

No idea what these two means or why these matters.